PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

LONGIDO DISTRICT JOINT EXAMINATION
FORM THREE

031/1 PHYSICS

TIME: 2:00 HRS APRIL 2019

MARKING SCHEME

SECTION A (30 MARKS)

1. @ 1 mark = 10 marks

Item no.	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answer	C	C	D	A	D	A	A	B	C	B

2. @ 1 mark = 10 marks

Item no.	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answer	G	D	M	A	I	L	B	E	H	C

3. @ 1 mark = 10 marks

(i) Velocity ratio
(ii) Inelastic collision
(iii) Relative motion
(iv) Parallax
(v) Hot air balloon
(vi) Luminous
(vii) Parallel arrangement
(viii) Vector
(ix) Limiting friction
(x) Windmill

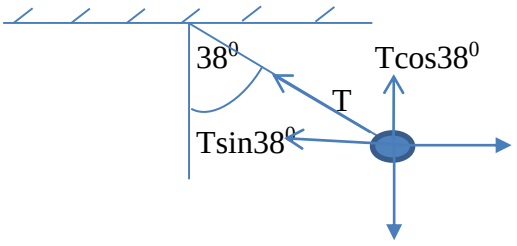
SECTION B (50 MARKS)

4. (a) **@ 1marks = 2marks**

(i) To make it float upright
(ii) To allow greater changes in height for every small change in density

(b) (i)Parallelogram law states that '' If two inclined forces are represented in magnitude and direction by the adjacent sides a parallelogram, their resultant is represented in magnitude and direction by the diagonal of the parallelogram passing through the point on intersection of the two sides'' **(3marks)**

(ii) Refer to the diagram below; **(5 marks)**



F **(1mark)**

W

Horizontal force = $T\sin 38^0$ **(1 mark)**

But $T\cos 38 = W = 60N$ **(1 mark)**

$T = \frac{60^0}{\cos 38} = \frac{60}{0.7880} = 76.14N$ **(1 mark)**

Therefore Horizontal force, $F = Tsin38 = 76.14xsin\ 38 = 76.14x\ 0.6157 = 46.88N$ **(1 mark)**

5. (a) (i) **@ 1mark for any two points = 2 marks**

- When used in walking and movement
- When used in moving body to stop
- When used to wear unneeded layers of some material
- When used in lighting match stick

(ii) Disadvantages of friction; **@ 1 mark for any two points = 2 marks**

- causes wear and tear
- loss of heat energy
- slows down motion of the body
- heat can cause appliance to burn
- it causes wounding, when skin wearing

(b) (i) friction is the force which resists a body from motion. **0.5 marks**

(ii) Laws of Friction **@ 0.5 marks= 2 marks**

- Friction depends on the nature of surface in contacts.
- Friction does not depend on the surface areas in contact.
- The ratio of friction force and normal reaction is equal to a constant value known as coefficient of friction
- Frictional force is directly proportional to the normal force between the two surfaces in contact.

(c)(i) The attraction force between two bodies can raise friction where particle from that object tend to bind each other so it is difficult to break that bond. **O.5 marks**

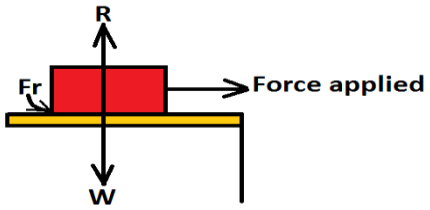
(ii) Given

Mass, m = 3kg

Force applied, F = 20N **(0.5 marks)**

Coefficient of kinetic friction, $\mu_k = 0.5$

Refer to the diagram below;



Using second Newton's law of motion; Net force = ma

But Net force = Applied force, F –Frictional force

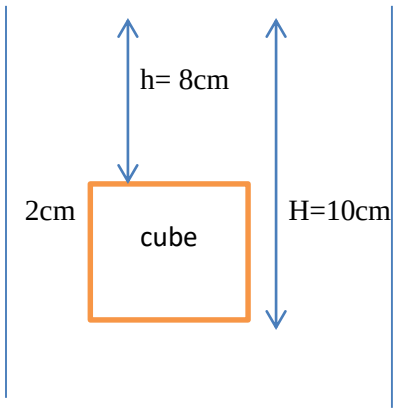
Net force $= F - F_r = F - \mu_k R = F - \mu_k mg$ **(3 marks)**
Then;
$F - \mu_k mg = ma$
$a = \frac{F - \mu_k mg}{m} = \frac{20 - (0.5x3x10)}{3} = 1.67 m/s^2$
$\therefore \boldsymbol{a = 1.67 m/s^2}$

6. (a) **@ 0.5 marks = 1.5 marks**
(i) Newton is the force required to give a body of mass 1kg with an acceleration of 1m/s^2
(ii) Inertia is the ability of a body to resist changing its state of motion or rest.
(iii) Linear momentum is the product of mass of an object and its velocity.
(b)(i) **(2.5 marks)**
Let time taken by stone 1,$t_1 = t$
Time taken by stone 2, $t_2 = t - 2$
From $V_2 = U_2 - gt$
$10 = 20 - 10(t - 2)$
$10 = 20 - 10t + 20$
$10t = 40 - 10$
$t = \frac{30}{10} = 3s$
But $t_2 = t - 2 = 3 - 2 = 1$
$\therefore \boldsymbol{t_2 = 1s}$

(ii) **(2 marks)**
Velocity of the first stone, $V_1 = U_1 - gt_1$
$V_1 = 20 - 10x3 = -10m/s$
$\therefore V_1 = -10m/s$
The negative shows that the stones moved in opposite direction after collision
(c)Given
Mass M = 5kg
Mass of A, $m_A = 2kg$
Velocity of A, $V_A = 60m/s$
Mass of B, $m_B = 3kg$
(i) **(2 marks)**
Velocity of B, V_B =?
Using principle of conservation of linear momentum;
$Momentum\ of\ bomb = momentum\ of\ exploded\ parts$
$Mu = m_A V_A + m_B \boldsymbol{V_B}$
$5x0 = 2x60 + 3x(-V_B)$
$0 = 120 - 3V_B$
$V_B = \frac{120}{3} = 40m/s$
$\therefore V_B = 40m/s$
(ii) **(2 marks)**

Total kinetic energy, K.E $= K.E_A + K.E_B = \frac{1}{2} m_A {u_A}^2 + \frac{1}{2} m_B {u_B}^2$
$K.E = \frac{1}{2} x2x60^2 + \frac{1}{2} x3x40^2 = 3600 + 2400 = 6{,}000J$
$\therefore$ **Total kinetic energy, K.E = 6,000 J**

7. (a) Altimeter is an instrument used to measure the attitude. **(1 mark)**
(b**) @ 1.5 marks**
(i) This is because at high attitude the atmospheric pressure is low compared to that of the body. This high difference in pressure causes the hurt to pump blood at high pressure causing the smooth and soft blood vessels mostly in nose to burst hence causing nose bleeding
(ii) This is because pebbles have small surface area which offers greater pressure hence causing pain.
(c)Given
Sides of cube, L = 2cm
Depth of cube, H= 10cm
Diagrammatically;



(i) **(3 marks)**
Pressure at the top (P_t) is due to the height ***h.*** i.e. $P_t = \rho gh$

Pressure at the bottom ($P_{b)}$is due to the depth **H** i.e. $P_b = \rho gH$

Difference in pressure, $\Delta P = P_b - P_t = \rho g(H - h) = 1000x10(0.1 - 0.08) = 200N/m^2$

(ii) **(2 marks)**
Weight of displaced water $= m_w g = \rho_w V_w g$

But Volume of water, $V_w = V_w olume\ of\ cube = (0.02x0.02x0.02)m^3 = 8x10^{-6}m^3$

Therefore, Weight of displaced water$= 1000x8x10^{-6}x10 = 0.08N$

8. (a) **@ 1 mark**

$1^{st} Newton's law\ of\ motion\ states\ that''$ Everybody continues in its state of rest or uniform motion in a straight line provided no external force is acting on it''

$2^{nd} Newton's law\ of\ motion\ states\ that\ ''$ The rate of change of linear momentum of a body is direct proportional to the applied force and it takes place to the direction of force''

$3^{rd} Newton's law\ of\ motion\ states\ that\ ''$ To every action there is an equal and opposite reaction''

(b) **@ 1.5 marks = 3 marks**

Elastic collision	Inelastic collision
Kinetic energy and momentum are conserved	Kinetic energy and momentum not conserved
After collision the bodies do not stick together	After collision the bodies stick together
After collision the bodies move with different velocities	After collision the bodies move with common velocity

(c) **(4 marks)**

Given

Mass of A, $M_A = 8kg$
Mass of B, $M_B = 4kg$
Initial velocity of A, $u_A = 20m/s$
Initial velocity of B, $u_B = 10m/s$
Final velocity of A, $V_A = 15m/s$
Final velocity of B, V_B =?

Using principle of conservation of linear momentum;

$M_A u_A + M_B u_B = M_A V_A + M_B V_B$
$(8x20) + (4x10) = (8x15) + 4V_B$
$160 + 40 = 120 + 4V_B$
$200 - 120 = 4V_B$
$V_B = \frac{80}{4} = 20m/s$

$$\therefore \boldsymbol{V_B = 20m/s}$$

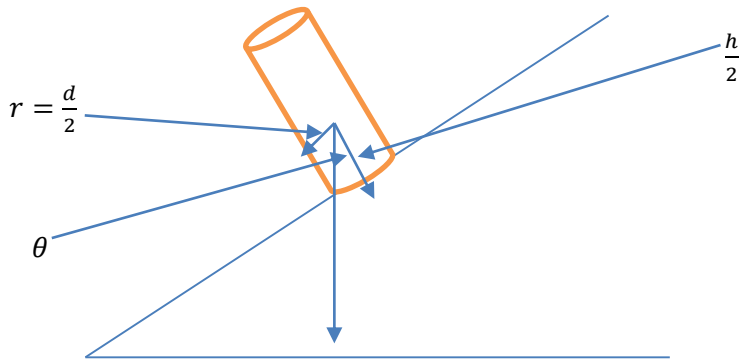
SECTION C (20 MARKS)

9. (a) Moment of force is the product of force and its perpendicular distance from a point to the line of action of the force. **2 marks**

(b) (i)Conditions for a body to be in equilibrium; **@ 1.5 marks = 3 marks**

- The base should be wide
- The center of gravity should be low

(ii) Diagramatically**,** **(5 marks)**



From the diagram above;

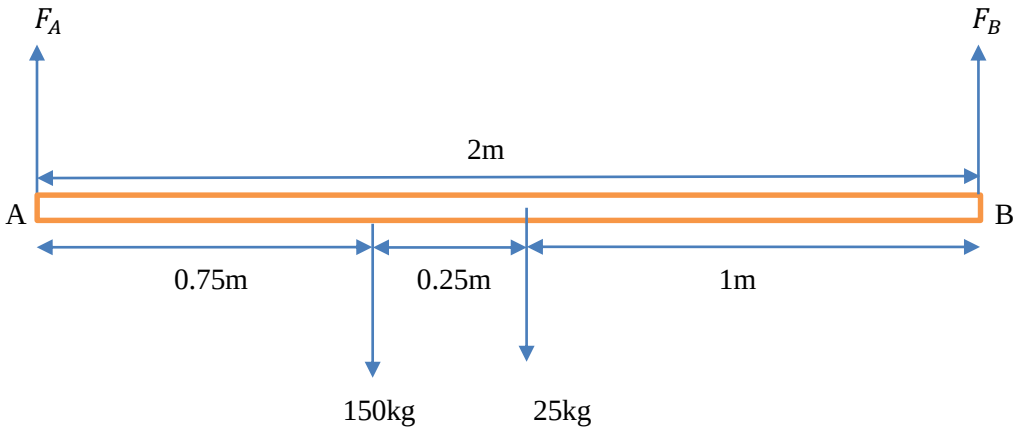
$$tan\,\theta = \frac{\frac{d}{2}}{\frac{h}{2}} = \frac{d}{h} = \frac{0.2}{0.4} = 0.5$$

$$\theta = tan^{-1}(0.5) = 26.6^0$$

$$\therefore\ \boldsymbol{\theta = 26.6^0}$$

(c) (i) Principle of moment states that "If the system is at equilibrium, sum of clockwise moment is equal to sum of anticlockwise moment'' **(2 marks)**

(ii) Diagramatically;



Using principle of moment;

Taking moment at B.

$(150x1.25) + (25x1) = 2F_A$ **(3 marks)**
$187.5 + 25 = 2F_A$

$$F_A = \frac{212.5}{2} = 106.25kg = 1062.5N$$

$$\therefore\ \boldsymbol{F_A = 1062.5N}$$

Solving for F_B

Sum of upward forces = sum of downward forces

$$F_A + F_B = (150 + 25)x10$$

$F_A + F_B = 1750 =$ **(3 marks)**
$F_B = 1750 - F_A = 1750 - 1062.5 = 687.5N$

$$\therefore F_B = 687.5N$$

(ii) Assumptions **@ 1 mark = 2 marks**

- The beam is assumed to be fixed at the extreme ends i.e at the exact end-tips of the beam
- The beam is rigid i.e it does not bend

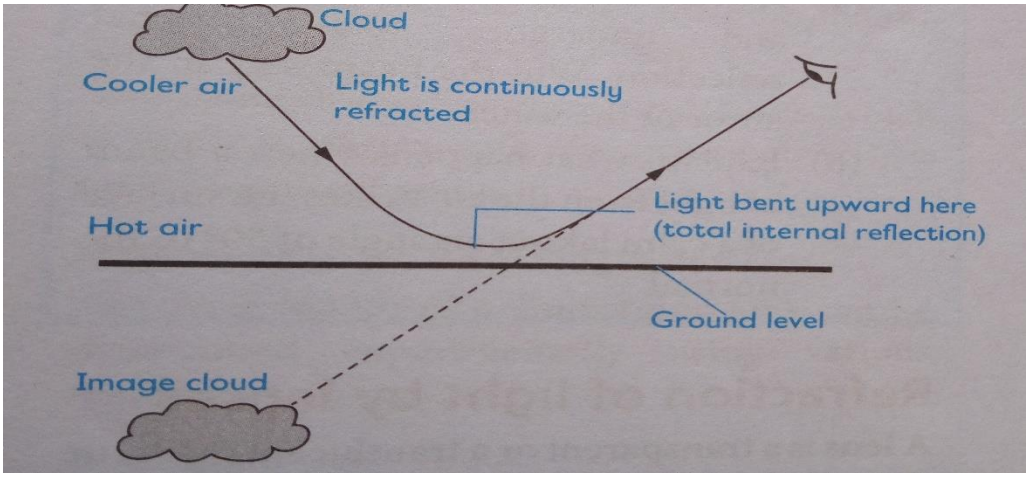
10. (a) Principles of ray diagram drawing; @ 1 mark = 3 marks

i) The ray of light parallel to the principle axis is reflected through the principle focus

(ii) The ray of light through the principle focus is reflected parallel to the principle axis

(iii) The ray of light through the centre of curvature is reflected back to its own path

(b) (i) Refer to the diagram below; **(4 marks)**



(ii) Mirage is formed as a result of continuous bending of light (refraction) as it passes through layers of hot air and cold air. As the light travels at an angle from the sky, it passes through layers of air decreasing refractive indices, consequently the light is gradually bent away from normal. At some point the angle of incidence exceeds the critical angle causing all light to be reflected upward. This looks like the reflection produced by a pool of water. **(3 marks)**

(c) (i) **(5 marks)**
Given

$$\mathbf{U = \left(\frac{1+m}{m}\right) f}$$

From magnification formula;

$$m = \frac{V}{U} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$$

Again using mirror formula;

$\frac{1}{f} = \frac{1}{U} + \frac{1}{V}$

$V = \frac{Uf}{U-f}$

Substituting the value of V into eqn (i) above;

$m = \frac{\frac{Uf}{U-f}}{U} = \frac{f}{U-f}$

$m(U-f) = f$

$mU - mf = f$

$mu = f + mf$

$U = \frac{f+mf}{m}$

$$\therefore \boldsymbol{U = f\left(\frac{1+m}{m}\right)}\textbf{, hence shown.}$$

(ii) Given

Focal length, f= 40cm

(a) 2.5 marks

Object distance, U=? if $H_i = 2H_o$

From $m = \frac{H_i}{H_0} - - - - - - - - - - (i)$

Again, $m = \frac{V}{U} - - - - - - - - - - - (ii)$

Equating eqn (i) and (ii) above;

$\frac{V}{U} = \frac{H_i}{H_0}$

$V = U\left(\frac{H_i}{H_0}\right) = U\left(\frac{2H_o}{H_o}\right) = 2U$

Substituting the value of V into $\frac{1}{f} = \frac{1}{U} + \frac{1}{V}$

$\frac{1}{40} = \frac{1}{U} + \frac{1}{2U} = \frac{2+1}{2U} = \frac{3}{2U}$

$2U = 120$

$U = \frac{120}{2} = 60cm$

$$\therefore \boldsymbol{U = 60cm}$$

(b) 2.5 $marks$

$H_i = \frac{H_0}{2}$

$Again\ from\ V = U\left(\frac{H_i}{H_0}\right) = U\left(\frac{\frac{H_0}{2}}{H_0}\right) = \frac{U}{2}$

$Substituting\ the\ value\ of\ V\ into\ \frac{1}{f} = \frac{1}{U} + \frac{1}{V}$

$$\frac{1}{40} = \frac{1}{U} + \frac{1}{\frac{U}{2}} = \frac{3}{U}$$

$$\therefore \boldsymbol{U = 120cm}$$